Department of Mathematics and Statistics
College of Science
Polytechnic University of the Philippines

Course Information

Course Code:	MATH 203	**Course Name**	Abstract Algebra 1
Units	3	**Prerequisite**	MATH 105
Course Description	This course introduces students to the fundamental concepts of modern algebra. Beginning with the basics of group theory, the course explores on standard algebraic concepts such as cosets, homomorphisms, and quotient structures. The latter portion of the course transitions into introductory ring theory, gaining exposure to rings, integral domains, and fields.		

Lecturer

Name	John Patrick B. Sta. Maria	**Office**	S514
E-mail	jpbstamaria@pup.edu.ph		

Course Plan

Week	Topic / Activity
1 – 2	Course Orientation
	Binary Operations and Modular Arithmetic
3 – 5	Groups and Subroups
	Group Isomorphisms
	Cyclic Groups
6	Review and Long Examination 1 mode: *in-person*
7 – 9	Permutation Groups
	Group Actions
	Dihedral Groups
10 – 11	Cosets and the Lagrange Theorem
	Normal Subgroups and Quotient Groups
12	Review and Long Examination 2 mode: *in-person*
13 – 14	Group Homomorphisms
	Group Isomorphism Theorems
	Finite Abelian Groups
15 – 16	Rings, Fields, and Integral Domains
	Ideals, Quotient Rings, and Ring Homomorphisms
17 – 18	Review and Long Examination 3 mode: *online*
	Final Examination mode: *in-person*

Grading System

(50 – base)

Rating Transmutation Function:

- $\text{Rating} = \left(\dfrac{\text{Raw Score}}{\text{Max Score}} \times 50\right) + 50$

Final Grade Components:

- Class Standing [CS]:
  - long exam 1 – 40%
  - long exam 2 – 40%
  - long exam 3 – 20%
- Major Exam [ME]:
  - final exam

Final Grade [FG]:

- $\text{FG} = (0.7 \times \text{CS}) + (0.3 \times \text{ME})$

Note: FG is rounded off to the nearest integer and is converted to the University's Point Grade System

Point Grade System

Final Grade	Point Grade
97 – 100	1.00
94 – 96	1.25
91 – 93	1.50
88 – 90	1.75
85 – 87	2.00
82 – 84	2.25
79 – 81	2.50
76 – 78	2.75
75	3.00
50 – 74	5.00
Withdrawn	W
Dropped	D
Incomplete	INC

References and Other Materials

1. **A First Course in Abstract Algebra**, John B. Fraleigh, 7th Edition
   - Click here while it's available.
2. **Abstract Algebra**, David S. Dummit and Richard M. Foote, 3rd Edition
   - Quickly paced, but comprehensive.
3. **Abstract Algebra: Theory and Applications**, Thomas W. Judson
   - Free e-book. Click here to download.

Department of Mathematics and Statistics
College of Science
Polytechnic University of the Philippines

Class Policies

1. **Active Participation**
   Attend and actively participate on every lecture. MATH 203 is highly cumulative. Don't make it anymore stressful by not exerting an effort. If one misses a lecture, you will like struggle with the future lessons.

2. **Focus**
   Avoid distractions during class lectures. Minimize the use of electronic gadgets to note taking and computing purposes only.

3. **Attendance**
   Attendance will be monitored, but not graded.

4. **Absence**
   Generally, you are allowed to have "bad days" on other days except on exam days. However, an absence during an exam requires **formal excuse**, provided that the reason is valid as deemed valid by the lecturer. Exams are typically planned and announced.

   - For **medical reasons**, secure medical certifications from healthcare professionals and have them validated/ authenticated by the University Health Services (UHS) before going to the Office of Counseling and Psychological Services (OCPS), who will issue the absence/admission slip.
   - For **non-medical reasons**, bring formal excuse letter to the (OCPS) with the necessary attachments for validation purposes.

5. **Missed Long Exam**
   In case of absence with formal excuse, re-taking the exam is not necessary. The rating for the missed long exam is set to be the minimum rating of the other two long exams.

6. **Missed Final Exam**
   As a short answer, you can't. However, if you must, an INC grade will be encoded in the SIS as your grade. Completion of grades can be made in the regular semester that follows. **Note:** An INC grade lapses to a final grade of 5.00 after a year of not completing the grade.

7. **Office Hours**
   I am usually available for queries and consultation on weekdays, from 8:00AM to 5:00PM. Practice professionalism. Use official channels for communication and use appropriate language, but do not overdo it by sending AI - generated templates.

8. **Class Representatives**
   They are not personal secretaries. Personal concerns must be directed to the lecturer by the student.

9. **Academic Integrity**
   In higher mathematics, collaboration is important. To balance learning with honesty, we impose the following rules.
   - **Discuss**. You are encouraged to brainstorm topics and problems, except during an exam.
   - **Write**. Once you move from discussion to writing your final submission, you must do so alone.
   - **Attribute**. If you worked significantly with a classmate, simply note their name in your work, if applicable.

10. **Technology and AI Policy**
    Using large language models (AI) to generate answers to assessments is a violation of PUP's policy on academic integrity. While the use of scientific calculators and mathematical softwares (like GeoGebra, etc) is encouraged, generative AI kills your growth as a student. Learning to spot the flaws of an AI model can be a good training, but submitting them as your own is a failure of your education.